

# ENTERPRISE CHAUFFEUR OPERATIONS PLATFORM

## COMPLETE SYSTEM MANUAL, FEATURE GUIDE & BUTTON DIRECTORY

**Architectural Mandate:** *ONE BOOKING → ONE RECORD → ONE SOURCE OF TRUTH*

This master documentation manual provides an exhaustive, granular directory of every visual button, user interface interaction, background mathematical algorithm, automated lifecycle state transition, and role-based operational gate across the entire Enterprise Chauffeur Platform (Backend Engine + 3D Luxury Frontend Dashboard Suite).

### 1. Global Navigation Bar & Top Header Controls

Located persistently across all pages to provide real-time telemetry, session controls, and navigation.

| UI Control / Button     | Type / Trigger                 | Functional Behavior & Core Logic                                                                                                                                                                                  |
|-------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobile Menu Hamburger   | Touch / Click Icon             | Opens slide-out mobile navigation drawer on smartphones with smooth backdrop blur. Auto-closes upon destination selection.                                                                                        |
| Melbourne Hub Telemetry | Live Real-Time Widget          | Displays active Australian Eastern Standard Time (AEST) with real-time second sweep, synchronizing Melbourne airport flight curfew gates.                                                                         |
| FastAPI 2.0 Engine Pill | System Status Badge            | Monitors async engine health, rate limiter status, database connection pools, and real-time backend latency.                                                                                                      |
| RBAC Persona Switcher   | Interactive Dropdown (6 Roles) | Dynamically hot-swaps active session between <b>Admin</b> , <b>Operations Lead</b> , <b>Live Dispatcher</b> , <b>Accountant</b> , <b>Driver</b> , and <b>Corporate Client</b> with instant permission re-scoping. |
| Notifications Bell (3)  | Alert Drawer Trigger           | Pulls high-priority operational alerts including flight delay reschedules, negative margin warnings, and pending balance chasing.                                                                                 |

### 2. Executive 3D Command Center (1)

The primary high-altitude dashboard providing operational oversight, financial KPIs, and 3D visualizers.

| Feature / Button            | Component          | Operational Impact & Details                                                                                                                  |
|-----------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Review Pending Queue (2)    | Glowing CTA Button | Instantly routes the operations dispatcher to the live dispatch operate board filtered directly to unallocated master journeys.               |
| Gross Revenue Card          | Hero Metric 1      | Calculates real-time gross passenger revenue (\$ AUD) inclusive of 10% Australian GST, with automatic Ex-GST net revenue breakdown.           |
| Net Operating Profit Card   | Hero Metric 2      | Calculates Net Operating Profit = Gross Revenue (Ex GST) - Direct Fleet Costs (Driver Payouts + Partner Offloads), displaying exact margin %. |
| Pending Dispatch Queue Card | Hero Metric 3      | Monitors unallocated master bookings awaiting human verification or driver conflict resolution with pulsating warning indicator.              |

|                           |                          |                                                                                                                                                         |
|---------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| On-Time Arrival Rate Card | Hero Metric 4            | Aggregates chauffeur punctuality (% of trips where arrived_at <= pickup_datetime) and fleet average passenger rating.                                   |
| 3D Luxury Car Showroom    | Three.js WebGL Canvas    | Interactive 360° mouse drag & touch orbit rotating luxury vehicle showroom with metallic clearcoat shaders, xenon lights, and alloy rims.               |
| Color Palette Selector    | Paint Picker (4 Colors)  | Dynamically swaps 3D car paint shaders between Obsidian Black (#0D1117), Champagne Gold (#D4AF37), Pearl White (#F8FAFC), and Royal Sapphire (#0F274A). |
| Fleet Class Selector Tabs | 5 Chassis Buttons        | Swaps 3D geometry and specs between Executive Sedan, Premium Sedan, Premium SUV, People Mover Van, and Minibus Shuttle.                                 |
| 3D Dispatch Radar Globe   | Three.js Airspace Canvas | Rotating 3D wireframe globe displaying flight trajectory arcs (MEL, SYD, BNE, PER), airport GPS pins, and active in-bound flight counts.                |
| Allocate Driver → (Queue) | 1-Click Quick Action     | Directly opens chauffeur allocation drawer for the selected master booking item in the Human-in-the-Loop queue.                                         |

### 3. Live Operate & Dispatch Board (`/operate`)

The real-time operational engine implementing the Add-Allocate-Settle lifecycle.

| Feature / Control            | Component                   | Functional Behavior & Safeguards                                                                                               |
|------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Table / Kanban Toggle        | View Switcher Tabs          | Switches between high-density operational Master Table View and Kanban Stage Board (PENDING, ALLOCATED, EN_ROUTE, COMPLETED).  |
| Search & Filter Bar          | Input Query Engine          | Instant multi-field filtering across Booking Number (`CCM-XXXX`), Passenger Name, Phone, and Route Address.                    |
| Allocate Chauffeur Modal     | Action Modal Trigger        | Opens allocation panel to assign driver and company vehicle. Automatically sets driver fixed payout allocation cost.           |
| Schedule Conflict Guard      | Automated Validation Engine | Prevents double-booking drivers by checking overlapping active jobs within a <b>90-minute safety buffer</b> .                  |
| Subcontractor Offload Drawer | Partner Dispatch Lane       | Broadcasts job offers to affiliate partners with a <b>15-minute countdown expiry window</b> . Enforces Negative Margin Guards. |

### 4. Instant 3D Quote & Multi-Leg Booking Engine (`/quotes`)

Customer and corporate quotation builder supporting multi-leg and hourly hire journeys.

**Key Features & Interactive Buttons:**

- **Journey Mode Selector:** Toggle between *Point-to-Point (One-Way)*, *Return Journey* (auto-applies 10% round-trip discount), and *Hourly As-Directed*.
- **Airport Meet & Greet Toggle:** Enables commercial flight number entry, auto-connecting the leg to real-time flight tracking.
- **10% Australian GST Calculator:** Auto-calculates 1/11th GST tax breakdown in real-time.
- **25% Deposit vs 100% Full Payment:** Split payment selector for advance bookings (> 7 days).
- **Confirm Master Booking (`CCM-XXXXX`) Button:** Creates master booking, child journey legs, and triggers confetti animation.

## 5. Driver Mobile PWA Portal (`/driver-portal`)

Dedicated smartphone interface for professional chauffeurs.

### Chauffeur 4-Step Trip Stepper Buttons:

- **1. En Route Button:** Driver starts travel to pickup. Updates leg status to EN\_ROUTE and sets driver to ON\_TRIP.
- **2. Arrived Button:** Driver reaches location. Updates status to ARRIVED and auto-dispatches passenger arrival SMS.
- **3. Picked Up Button:** Passenger boards the vehicle. Updates status to PICKED\_UP.
- **4. Completed Button:** Trip finishes. Leg marked COMPLETED, driver auto-toggled to AVAILABLE, and trip earnings added.
- **Privacy Isolation Gate:** Customer total fare is completely hidden from the driver; only driver payout is displayed.
- **Simulate GPS Telemetry Ping:** Sends live latitude/longitude coordinates to fleet dispatch map.

## 6. Airport Flight Radar & Automation (`/flights`)

FlightAware integration for flight monitoring and delay management.

### Automation Controls & Features:

- **Simulate Flight Radar Check Button:** Queries commercial flight tracking for real-time delays.
- **Automated Delay Rescheduler:** Automatically applies +30 min domestic and +45 min international customs buffers to pickup times upon detected delays.
- **Complimentary 60-Min Wait-Time Calculator:** First 60 minutes from touchdown are free; slider calculates \$1.50/min excess wait charges thereafter.

## 7. GST Invoicing, Tax & Remittance Engine (`/invoicing`)

Australian Tax Office (ATO) compliant financial and invoicing suite.

### Financial Controls & Invoicing Tools:

- **Tax Invoice Ledger (`INV-YYYY-XXXX`):** Itemized invoices with 10% GST breakdown and printable PDF modal.
- **FIFO Debt Allocation Remittance Tool:** Allocates corporate customer lump-sum payments to the oldest overdue invoice first.
- **Australian BAS Report View:** Summarizes quarterly GST Box G1 (Total Sales), Box 1A (GST on Sales), and Net Sales Ex GST.
- **Driver RCTI Settlement Generator:** Generates Recipient Created Tax Invoices with ABN GST registration credits.

## 8. Profit Analytics & CSV Exports (`/analytics`)

Executive margins, fleet utilization, driver KPI scorecards, and streaming CSV data exports.

### Analytics Visualizations & Export Buttons:

- **Trip Profitability CSV Export Button:** Generates RFC 4180 compliant CSV stream of per-trip gross fares, driver payouts, partner offloads, and net margins.
- **Financial Ledger CSV Export Button:** Exports full financial ledger of all credit/debit transactions and Stripe settlements.
- **Interactive Revenue vs Cost Bar Chart:** Hardware-accelerated SVG chart with hover tooltips for daily revenue, direct fleet costs, and net operating profit.
- **Chauffeur Performance Scorecard:** Real-time leaderboard tracking customer ratings, completed trips, and on-time arrival rate %.

## 9. Subcontractor Partner Network & Vehicle Fleet (`/partners-fleet`)

Subcontractor compliance monitoring and active company vehicle registry.

### Compliance & Fleet Controls:

- **Partner Insurance Compliance Gate:** Blocks job offloads to subcontractors whose Public Liability Insurance has expired.
- **Register Subcontractor Modal:** Adds partner with accreditation number, insurance expiry date, and contact details.
- **Fleet Vehicle Catalog:** Registers company vehicles with make, model, year, registration plate, and strict passenger/luggage capacities.